

A systematic review of methodologies for human behavior modelling and routing optimization in large-scale evacuation planning

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ABSTRACT

Frequent and escalating natural disasters pose an increasing threat to society and the environment. Effective disaster management strategies are crucial to mitigate their impact. This paper reviews recent methodologies for large-scale evacuation planning, a key element in risk reduction. A systematic analysis of 100 articles and conference proceedings in evacuation planning, focusing on human factors/behavior modeling and evacuation routing optimization, reveals that Agent-Based Simulation (ABS) is commonly used to predict human factors/behaviors. Heuristics/meta-heuristics and traffic assignment techniques dominate evacuation routing planning, often aiming to identify the shortest evacuation path. While evacuation decisions and route choice are extensively studied, optimization approaches frequently lack integration with human factors/behavior modeling. This review underscores the need for further research to enhance evacuation planning by integrating human factors/behavior and optimization methodologies for increased effectiveness and efficiency.

1. Introduction

The International Federation of Red Cross and Red Crescent Societies (IFRC) [1] defines a disaster as a natural or human-made phenomenon, such as flooding, hurricanes, earthquakes, tsunamis (natural), terrorist attacks, and toxic chemical leakage (human-made), that disrupts society's functioning beyond its capability to manage. The IFRC [1] also defines disaster management as the organization and coordination of resources to address the humanitarian aspects of emergencies, including mitigation, preparedness, response, and recovery. The occurrence of disasters has increased significantly globally in recent decades, with 7348 disasters being recorded between 2000 and 2019, marking a 73 % increase compared to 1980–1999 [2]. Geophysical catastrophes have resulted in numerous fatalities, displacement of residents, and destruction of infrastructure worldwide [3].

Evidence suggests that evacuation planning and management are crucial for mitigating risks during disasters. A bibliometric review [4] highlighted the steady growth in disaster evacuation literature since 2015, indicating the importance researchers place on such studies. Evacuation is a significant aspect of disaster management [5], involving moving individuals from disaster zones to secure areas [6,7]. Evacuation can be classified as small or large, no-notice or short-notice [8–10], and as compulsory, advised, or vol-

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untary [11–13]. Factors like warning time, response time, evacuation orders, diffusion process, routing, traffic network status, and transportation control measures greatly influence evacuation efficiency [14,15].

Recent work has shown that stakeholders, including governmental authorities responsible for disaster management, may encourage resilience, coordination, and effective disaster response and recovery procedures by identifying and evaluating risks and putting preventive measures and emergency plans in place [16]. For example, evacuation design helps authorities identify crowd bottlenecks and plan countermeasures in advance, maximizing successful outcomes. Effective evacuation processes require analyzing factors such as hazard type, network vulnerability, shelter parameters, and human behavior to designate transportation and evacuation routes. Simulating traffic networks helps interpret evacuation situations and the impact of rules and control measures [17]. Evacuation planning is an interdisciplinary problem involving engineering, operations research, traffic management, and social sciences [17,18], aiming to establish an integrated, optimized evacuation model [19]. However, incorporating realism in planning comes with a trade-off between model complexity and simplification for efficient parameter simulation. Incorporating network uncertainties, such as evacuation demand, road capacities, and evacuees' behavior is recommended [20], although the latter is often overlooked.

This paper presents a review of the methodologies used in evacuation planning. Although other reviews have been implemented on similar topics (Table 1), the work presented in this paper differs by investigating the evacuation planning methodologies with a view to combining both behavioral studies with optimization approaches, to improve the accuracy of evacuation models' outcomes. This approach was supported by prior review papers, which often posit that the involvement of human behaviors in designing evacuation plans needs more attention [3,17,21–28]. Thus, this paper aims to demonstrate the importance of, and methodologies for, integrating aspects of human factors/behavior in evacuation planning of emergencies in the context of large-scale natural and human-made disasters. The paper also presents optimization/simulation techniques that can solve vehicle/pedestrian routing problems, to check the possibility of charting human behaviors as constraints. The systematic review covers recent literature that evaluates large-scale residents' evacuation, including pedestrian and vehicular evacuation from cities, metropolitan areas, and large facilities, to identify suitable methodologies that can be utilized in evacuation planning.

2. Methodology

Fig. 1 displays a summary of the screening procedure to select the reviewed articles as well as the utilized data sets. We used Scopus (as suggested by Ref. [37]) and Google Scholar to cover a broader range across disciplines. We used "evacuation planning," "emergency," "optimization," "disaster," "human behavior," and "prediction" as the search keywords. The timeframe was set to span January 2012 and April 2024 to focus on recent and advanced methods. After filtering, 118 papers were examined for human behavior prediction methodologies among which 93 papers were selected as addressing optimization/simulation approaches. Further review of the evacuation problem scale, to focus only on large-scale disasters, brought the total number of selected articles to 85.

To ensure comprehensive coverage of relevant documents, we conducted an additional search using UK spelling terms (e.g., "optimization" and "human behavior"), which resulted in 12 additional papers. A further three articles [12,15,32] not initially captured during the search process, were also included based on their importance highlighted in a bibliometric study by Liu et al. [4]. As a result, in total, 100 papers were included in this systematic review – the full list of papers can be seen in Tables 2, 4 and 5 in the following sections.

Most of the selected publications were peer-reviewed, with a few preprint publications among them. The reviewed literature covered various fields such as safety, engineering, risk management, transportation management, computer science, operations research, and technology. Publications were included in the analysis only if they addressed large-scale evacuation planning methodologies concerning human factors/behavior or the optimization of evacuation routing approaches at microscopic, macroscopic, or mesoscopic levels in dynamic, stochastic, and robust evacuation modeling. The review excluded articles that explicitly represent building evacuations, except for studying crowd behavior at exits, examining the effect of a multi-story facility on evacuation efficiency, and large facilities (such as festival halls, educational buildings, city arenas, and industrial plants were included), under the focus on major disasters. All disaster contexts were included.

3. Evacuation planning methodologies

Papers were categorized based on their focus on human factors/behavior, evacuation routing optimization, or a hybrid. Further classification is considered the primary methodology in the evacuation planning process. Studies using multiple methods were categorized by the dominant one. Human factors/behavior prediction methods were categorized by hazard type/scale, and the method used and explored factors/behaviors. Evacuation route optimization was classified as the main method for route identification, covering various modes and considerations. Details such as hazard type/scale, model objective, decision/affecting variables, method, and case study were determined. The interdisciplinary literature integration review considered criteria from both human behavior modeling and optimization approaches.

3.1. Human behavior prediction/Modelling methodologies

Table 2 presents the summary details of the reviewed articles relevant to human factor/behavior prediction/modeling.

As shown in Table 2, various methods have been applied to predict/model a variety of human factors/behaviors. Table 3 illustrates the frequency of methods used in examining human factors/behaviors and the corresponding papers.

Methods were arranged (Table 3) in descending order according to their frequency in the literature. ABS is the highest usage frequency employed to model human behavior. Further details for each method are discussed in the following sections.

Table 1

Existing literature-based studies on the field of evacuation planning, and their shortfalls.

Ref.	Review Themes	Study aim/focus	Shortfalls
[3]	Earthquake Operations Management (OM) phases (mitigation, preparedness, response, and recovery)	Addresses the main approaches towards operations research in earthquake disaster management	Focus is limited to operations research role in emergency management
[4]	Literature distribution characteristics, citation analysis, and keyword occurrence analysis	Identify research hotspots and trends in the emergency evacuation discipline	Limited to a bibliometric review
[15]	Background data collection, trip generation, departure timing, and destination and route choice	Demonstrate the behavioral parameters that affect evacuation time in the context of a hurricane	Limited to a single evacuation context (hurricane)
[17]	Evacuees' behavior modeling, travel demand modeling, traffic assignment modeling, and trip distribution modeling	Highlights the past and present evacuation traffic simulation modeling and the consensus on evacuation behaviors	A general overview rather than an in-depth review of evacuation prediction methodologies
[21]	Traffic assignment methods, evacuation modeling goals, evacuation behavior evaluation, deterministic methodologies, evacuation dynamics, shelter allocation, and multi-model and mass-transit evacuation	Review studies that address supply and demand management techniques	The focus is limited to optimization approaches rather than considering both simulation and optimization. Human Factors did not receive specific consideration.
[29]	Evacuation process model (hazard, decision to evacuate, warning, withdrawal including evacuation behavior and routing, shelter, and re-entry process) regarding the transportation management	Highlights the recent developments in evacuation research concerning human behaviors, risk assessment, and transportation	The context of evacuation is limited to flooding events. This work is limited to the withdrawal stage only
[30]	The methodologies employed in investigating a specific type of disaster, in addition to the existing resource demand forecasting approaches	To identify the most suitable forecasting technique and decision variables under the disaster type and address studies that are concerned with statistical analysis and recent artificial intelligence techniques	Focus is restricted to traditional statistics, machine learning, natural-network-based, and simulation methods.
[22]	Pre-disaster functions (facility location), post-disaster functions (relief distribution and victims transportation), and further functions	Evaluating optimization models employed in solving emergency logistics problems	It was limited to optimization approaches that are concerned with emergency logistics.
[23]	Evacuation theories, modeling, decision-making, risk evaluation	Review the available literature concerning large-scale crowd evacuation	A broader review includes theories, modeling, decision-making, and risk assessment aspects, which do not provide detail on the specific methodological approaches.
[24]	Classification of the factors that impact the evacuation phase and levels of decision-making	Identify the impacting factors associated with evacuation planning	No contribution to the methodologies
[31]	Toxic-gas-leakage, fire, and certain visibility evacuation	Enhance the research procedure in chemical-concentrated area evacuation	Limited to a specific context of disaster (toxic gas leakage and fire)
[25]	Review of different crowds model types.	Investigation of crowd evacuation models	No methods for evacuation routing optimization/simulation have been reviewed.
[26]	Travel demand phase (trip-based and activity-based) and traffic assignment phase (static and dynamic methodologies)	Determining the most appropriate approaches to be employed for a wildland-urban interface fire traffic evacuation model in each stage	The context of evacuation is limited to wildland/urban interface fires. A specific focus on travel demand and traffic assignment
[27]	Objective, type of catastrophe, evacuation zone, simulation software, agents' attributes, contribution to evacuation planning, and data validation	Identify the challenges that come along with researching ABS to simulate pedestrian evacuation together with determining research similarities/differences	Review is limited to papers that involved ABS only in their research
[28]	The role of social science, computational social science, crisis informatics, and computational social science in disasters	Bridge the gap between the data science and the social discipline	Concentrated focal point on the technology development
[32]	Zoning, demand modeling, traffic assignment and routing preferences, evacuation strategies, and other related factors and logistics	A review of highway-based evacuation modeling and simulation	Only focusing on highway-based evacuation modeling and simulation. No account for pedestrians' evacuation. Hurricane events only.
[33]	Methodologies to assess human behavior are categorized based on evacuees' behavior regarding individual and crowd perspectives	Evaluate and classify the methodologies that have been presented up to date to create a more consistent approach to interpreting human behavior during an evacuation process	The main goal is to introduce and categorize the proposed methodologies, with no consideration for the actual influence of human behaviors on evacuation outcomes
[34]	The research methods and data sources; the findings of the literature analysis, comprising the number of articles published and the year of publication; the distribution of subjects, authors, and journals; the analysis of citations and co-occurrences of keywords; and the research hotspots of safety evacuation in China	Examining safety evacuation studies in the Web of Science core database from 1990 to 2021 using a combination of traditional and bibliometric literature review approaches. Its goal was to evaluate both the present and upcoming developments in this area.	A single focus on the research status of crowd evacuation published by Chinese scholars in the Web of Science

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Table 1 (continued)

Ref.	Review Themes	Study aim/focus	Shortfalls
[35]	Techniques for gathering data on human behavior from large-scale outdoor fire incidents; the body of empirical literature; and evacuation models that may be employed to predict how people would behave in different large-scale outdoor fire disasters	Understand the general idea of human behavior in large outdoor fires.	The context of evacuation is limited to outdoor fires. No focus on human behavior to protect properties
[36]	Gathering and extracting data, analyzing behaviors and phenomena, simulating and modeling, and applications and solutions.	Establish a survey framework to carry out a comprehensive study of subgroup research	A concentrated focal point on subgroup research in pedestrian and evacuation dynamics

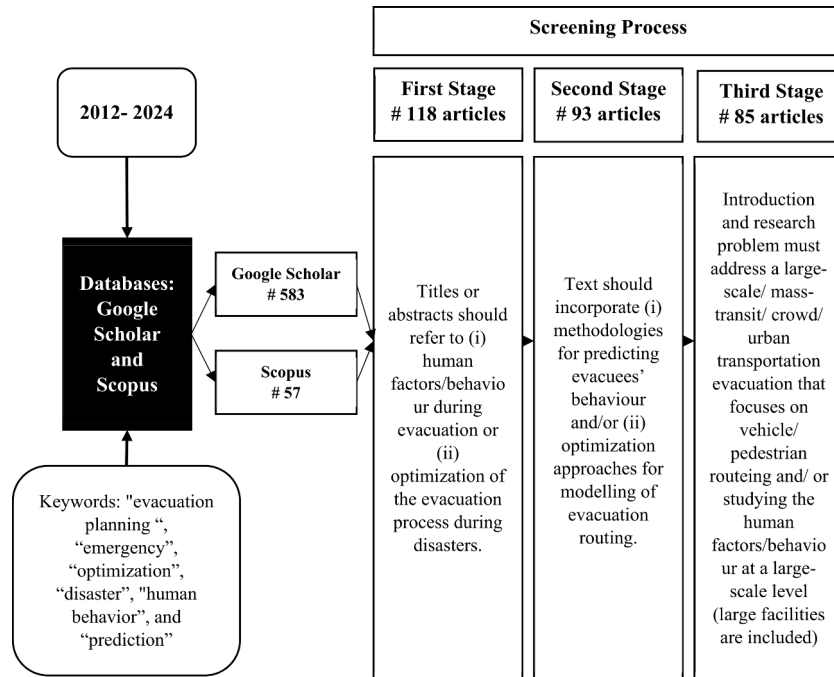


Fig. 1. The systematic review search criteria and screening process.

3.1.1. Agent-Based Simulation (ABS)

ABS is one of the most useful techniques for predicting human (agent) behavior by representing each evacuee as an agent, that is, as an entity with the ability to move freely and make decisions on their own using the information they collect from observing their surroundings and the set of rules that are specific to each kind of agent [39]. It is also a reliable method for representing different decisions by agents in different environmental situations. Furthermore, each agent (e.g. evacuees or emergency respondents) can be analyzed separately, which reflects a more realistic and accurate simulation. ABS can replicate autonomous environmental behaviors caused by natural disasters. This method recorded the highest frequency of usage compared to other methods.

ABS was employed to investigate a wide range of human factors/behaviors during emergencies, including but not limited to, the evacuation decision [38,40], demographics [39], familiarity [50], risk perception [51], social interaction [54], and vulnerability [55]. This indicates its great capabilities of addressing a variety of factors/behaviors, however, examining a limited number of human factors/behaviors is suggested to reduce modeling complexity. Empirical evidence can be obtained by comparing the raised behavioral patterns with real-world data.

3.1.2. Social Force Model (SFM)

SFM assumes that the crowd has driving forces (goals), interactions between agents, and interaction with the surroundings, in which they update their locations frequently to reach their targeted destination. Research using SFM has focused on crowd evacuation with consideration given to human behaviors such as exit choice [43,48], social interaction [44,64], and pedestrian dynamics [79]. Additional techniques are sometimes incorporated with SFM to enhance its capabilities, for example, the logit model [43], reinforcement learning [44], and the active rotation torque [48]. This indicates a drawback of SFM as it needs further methods to achieve adequate performance.

Table 2

Human factors/behavior prediction methodologies presented in the literature.

Ref.	Hazard/Evacuation/Scale	Method	Human Factors/Behaviors Aspects	
			Software	Factors/Behaviors
[38]	Hurricane	ABS/decision field theory	AnyLogic	The choice to evacuate, evacuees' perceptions of environmental changes, social interactions, and demographics
[39]	Seismic/large-scale/pedestrian	ABS	Unity 3D game engine	Age, gender, emotions, health state, fatigue, and the presence of debris influence walking speed and evacuation choices
[40]	Multimodal near-field tsunami	ABS	NetLogo	Evacuation options (stay in place, horizontally on feet, horizontally by car, vertically), departure time (immediately or delayed), and speed
[41]	Post-earthquake/crowd/pedestrian	CA	N/A	Familiarity, habitual preferences (walking and route choices), and protocol robustness (change resistance)
[42]	Vertical/dam-break flood	Optimization/the classical facility location problem	N/A	Heterogeneous walking speed and vulnerability (disabled and elderly individuals) together with evacuees' locations
[43]	Microscopic pedestrian/cell-based/crowd	SFM/discrete-choice model	AnyLogic	Exit choice
[44]	Large single-story construction area/crowd	RL/SFM/A* algorithm/proximal policy optimization algorithm	Unity 3D game engine	Social interaction with surroundings, walking behavior, cognitive stress
[45]	Mass/Lecturing hall/crowd evacuation	ABS	MASON, Java-based	Evacuation speed considering panic behavior
[46]	Major train station	Ordered logit model (statistical analysis)/survey	N/A	Demographics (gender, age), route/exit choice, herding, reactivity, and competitiveness
[47]	Festival event	Multi-agent simulation model	Scenargie	The selfish behavior of healthy individuals (not following evacuation instructions) which affects the disaster weak and walking speed
[48]	Pedestrian/several door widths	The three-circle SFM	MPU-6050 device	Competitive behavior influences rotation dynamics near bottlenecks at exits
[49]	Urban transportation/aftermath nuclear	Disaggregated ABS	N/A	Herding behavior, travel mode choice (walking or accessibility to vehicle), household recognition (individually or as a unit), panic, shelter & healthcare seeking, and help & assist
[50]	Urban/Chlorine spill evacuation	ABS	NetLogo, ALOHA, HEEDS, and MO-SHERPA	Knowledge (familiarity with the city), awareness (risk perception), and sense of urgency (evacuee's stress level)
[51]	Large-scale/volcanic	ABS	Anylogic Personal Learning Edition	The decision to evacuate, risk perception, social interaction, evacuating speed, destination choice, demographics (gender, age, and education level)
[52]	Vertical near-field tsunami evacuation	ABTEM	NetLogo	Milling time, walking speed, and the choice to evacuate
[53]	Large-scale wildfire	WDM/HCM	Glm function in R	Risk perception influenced by (age, education, income, gender, observed cues, and house-staying period) varies between (natural, information seeker, cautious, and defensive), the choice to evacuate, and departure time
[54]	Homogenous groups/pedestrian/crowd evacuation in an indoor public facility	ABS	NEAT	Knowledge, updated experience, social interaction with other evacuees and the environment, obstacle avoidance behavior, and route choice
[55]	Hurricane	ABS/decision field theory	N/A	Evacuees' vulnerabilities based on demographics (gender and age) and socioeconomics (income level, family or individual member, house ownership, and education)
[56]	Tsunami	Questionnaire	R/SPSS	Evacuation intention

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Table 2 (continued)

Ref.	Hazard/Evacuation/Scale	Method	Human Factors/Behaviors Aspects	
			Software	Factors/Behaviors
[57]	Volcanic	ABS/(threshold-based decision rule depending on the normal-investigating-evacuating state model)	AnyLogic	The decision to evacuate
[58]	Pedestrian/crowd	Multi-attribute binary discrete-choice model	N/A	Exit choice adaptability is influenced by (congestion at exits, exit visibility, social interaction, and inertia)
[59]	Hurricane	ABS	FLEE	The decision to evacuate is influenced by risk perception (forecast, evacuation instructions, demographics & socioeconomics) and information flow
[60]	Terrorist attack	Manual approach (observing videotapes) and statistical analysis	Google Maps/Street View/MATLAB	Evacuation speed and evacuees' movement
[61]	Large event halls	ABS	NetLogo	Walking speed, age, mobility disabilities, familiarity with the surroundings, grouping behavior, nearest egress selection, routing choice
[62]	Mass/urban/technological risk/flood	ABS	ESCAPE with the use of BEN is based on BDI	Emotional state, emotional infection, personal characteristics, relationships, and norms
[63]	Pedestrian/urban tunnel/fire	A field experiment	N/A	Routing choice, evacuation time, bodily status and response to tunnel technical structure, gender, age, and emotional state
[64]	Outdoor/flood	SFM	Setup-based generic software/FloPEDS/MassMotion	Evacuees' movement (speed) and interaction with the surrounding environment and each other
[65]	Large-scale	ABS/BDI system	JACK/IC/LT	The influence of the diffusion process (spread of knowledge, speculations, and decisions) on the evacuees' psychology
[66]	Hurricane	ABS	N/A	Shelter choice with regard to the evacuees' relationships
[67]	Urban residential communities	Survey	Pathfinder	Demographics (gender, age, education, bodily state, family size, knowledge), initial response and pre-evacuation time estimation, and routing choice
		Random forest algorithm		Pre-evacuation time influenced by pre-evacuation behaviors obtained from surveys
[68]	Wildfire	Rule-based algorithm	N/A	Evacuation decision, compliance rate, and departure time
[69]	All contexts	FKNN/Holt-Winters	R studio	Panic
[70]	Flood	ABS	NetLogo	Evacuation behavior (walking/mobility) of older people
[71]	Tsunami	ABS	Artisoc 4.0	Departure time, evacuees' speed, and route choice (random, shortest, and herding)
[72]	Mass/tsunami	ABS/SFM	C++/OpenMP	Evacuees' movements and social interaction
[73]	Large-scale	ABS/latent class multi-nominal logit model	MATSim	Destination choice, mode choice, evacuation decision, and socio-demographics
[74]	Crowd pre-evacuation decision	Questionnaire/probit regression model/utility functions	N/A	The decision to evacuate (including delays) is influenced by risk perception and herding behavior (the significance of age, gender, mode choice, zone, group, and time have been investigated as well)
[75]	Earthquake	Multi-agent simulation model/Bayesian inference method	NetLogo	Route choice (influenced by situational awareness), speed, age, movement disorder, familiarity with the surroundings, and awareness of the nearest exit
[76]	High-density community/earthquake	Multi-agent simulation model	ArcGIS/NetLogo	Psychological panic and cooperative behaviors

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Table 2 (continued)

Ref.	Hazard/Evacuation/Scale	Method	Human Factors/Behaviors Aspects	
			Software	Factors/Behaviors
[77]	Hurricane	Interpretable machine learning (enhanced logistic regression model)	N/A	Household-level evacuation decisions (influenced by demographic, geographic, and resource-related predictors)
[78]	Phased/hurricane	Latent class logit model	NLOGIT	Compliance with a phased hurricane evacuation order (influenced by stakeholder, risk, and evacuation perceptions, destination choice, number of vehicles, tenancy period, and travel-delay expectation)
[79]	Crowd	SFM/machine learning	MATLAB (Version R2022a)	Pedestrian dynamic (vulnerable pedestrians are those who move more slowly or are farther away from exits)
[80]	Hurricane	Machine learning (Zero-Inflated Extreme Gradient Boosting Tweedie Regressor (XGBoost))/smartphone location data	R/Spectus Data Cleanroom platform	Evacuation destination choice
[81]	Hurricane	The p-median and p-center optimization models	CPLEX for AMPL solver package	Nearest-shelter behavior
[82]	Tsunami/vertical	VR/questionnaire/GIS	3D Vista software/Tsunami-HySEA software/NVivo/ArcGIS	Vertical evacuation decision-making (influenced by evacuation experience, awareness, knowledge about earthquakes and tsunamis, and knowledge about emergency-response plans)

N/A- Not Applicable, ABS- Agent-Based Simulation, CA- Cellular Automata, SFM- Social Force Model, WDM- The Wildfire Decision Model, RL- Reinforcement Learning, ABTEM- Agent-Based Tsunami Evacuation Model, HCM- Hybrid Choice Model, NEAT- NeuroEvolution of Augmenting Topologies, FLEE- Forecasting Laboratory for Exploring the Evacuation-system, IC- Independent Cascade, LT- Linear Threshold, BDI- Belief-Desire-Intention, FKNN- Fuzzy K-Nearest Neighbor, VR-Virtual Reality, GIS- Geographic Information System.

Table 3

Ratios of methods used in investigating human behaviors in the literature.

Method	References
ABS (43 %) ^a	[38–40,45,49–52,54,55,57,59,61,62,65,66,70–73]
SFM (11 %) ^a	[43,44,48,64,79]
Survey/questionnaire (9 %) ^a	[46,56,60,74]
Computational modelling (7 %) ^a	[67,68,83]
Choice models (7 %) ^a	[53,78,84]
MAS (7 %) ^a	[47,75,76]
Machine learning (4 %) ^a	[77,80]
Optimization (4 %) ^a	[42,81]
Field experiment (2 %) ^a	[63]
CA (2 %) ^a	[41]
IoT (2 %) ^a	[69]
VR (2 %) ^a	[82]

^a Percentages have been rounded to the closest number.

3.1.3. Surveys/questionnaires

Investigating participants' thoughts or experiences to handle similar upcoming situations and/or establish effective countermeasures is a good strategy in evacuation planning. Acquiring data from individuals who have suffered major incidents can be difficult or even impossible, since many casualties may perish in the emergency, and often reports on deaths are confidential. Nevertheless, researchers like Bernardini and Quagliarini [60] reviewed tapes from the latest terrorist attacks around Europe, and later, conducted qualitative and quantitative assessments to support the development of more integrated risk response plans. Shiwakoti et al. [46] studied the potential behaviors (likelihood of their occurrence) undertaken by train passengers during disasters by evaluating their responses through a questionnaire distributed to 1134 participants. The results demonstrated that passengers tended to be reactive and waited until they were instructed by the authorized people; they were cooperative and helped those needing it, following each other in the evacuation. Another study [56] was conducted by displaying photographs representing different tsunami evacuation scenarios for Puerto Saavedra using a structured questionnaire. The findings demonstrate that populations tend to evacuate to sites away from the coastline, higher beyond sea level, and far from emergency infrastructure and evacuation routes. These conclusions can assist disaster management by establishing more realistic evacuation planning guidelines.

Table 4

Evacuation routing planning optimization methodologies presented in the literature.

Ref.	Hazard/Evacuation/Scale	Method	Optimization Aspects		
			Model objective	Decision/affecting Variables	Case Study
[85]	Medium-size urban area/flood	Traffic Assignment (Nearest Allocation)/Spatio-Temporal Optimization Model (STOM)	Minimize the total evacuation time	Accessibility, route capacity, shelters' capacities, evacuee's vulnerability, network vulnerability	Valley of Tours and Valley of Gien, France
[86]	Urban area	Traffic Assignment/User Equilibrium	Minimize the total evacuation time	Congestions caused by diversions and road closures	Central Business District (CBD), Washington/hypothetical
[87]	Urban area/terrorist attack or a natural calamity	Traffic Assignment/User Equilibrium	Maximize the highway network efficiency along a path k (evaluate the efficiency)	Congestions caused by diversions and road closures	Hypothetical case study
[88]	Hurricane	Column generation approach/Evacuation Decision Tree Analysis (EDTA)	Minimize the total flow time or the makespan	Network flow	Hurricane Irma (2017)
[89]	Toxic gas leak	Integer Programming (Time-Dependent Shortest Path Problem (TDSPP))/neural network surrogate model	Optimize evacuation routing by minimizing the total prospective risk of the evacuation	Location of the origin nodes, shelter nodes, evacuation speed, wind speed, wind direction, leakage rate of toxic gas, and leakage position	Mipo Industrial Complex located in Ulsan, South Korea
[12]	Large transportation network	Integer Programming and M/G/c/c queuing model	Identify a set of exit paths to move evacuees to safety within the transportation network by minimizing total traveled distance, total clearance time, and blocking	The assignment of evacuees to the shortest paths	A small hypothetical case study
[90]	Toxic gas release	A customized Dijkstra algorithm together with a minimum cost maximum flow algorithm	Optimization of evacuation routing and the optimal decision between shelter-in-place and evacuation trade-off (risk assessment and decision support)	Evacuation flow and health consequences	"3-29" liquid chlorine leakage accident in China
[91]	Tsunami	First-in-first-out (FIFO) queuing simulation using MATSim	Evaluate the variations in clearance times	Demand (evacuees) and supply (traffic network)	Balboa Island in the City of Newport Beach, California
[92]	Flood	Dynamic Traffic Assignment/Agent-based modeling	Evaluates planned evacuation before a flood	Self-evacuation by private vehicle transport	Warsaw, Poland
[93]	Hurricane	A scenario-based stochastic integer programming model	Minimizing the total expected evacuation costs from vehicle trips and the fixed cost from setting up one or more staging areas	Evacuation demand, transport vehicles with varying capacities, and both critical and non-critical patients	Hurricane Harvey of 2017
[94]	Flood	Classification, Dijkstra's, and Random Forest algorithms	Evaluating evacuation vulnerability and risk based on the predicted evacuation time	Vulnerability, time-varying flooding hazards, and estimated congestion levels	Joso City, Japan

Table 5

Methods used in solving evacuation routing problems in the literature.

Method	Research
Traffic Assignment (27 %) ^a	[85–87]
Integer Programming (IP) (27 %) ^a	[12,89,93]
Simulation-based Approaches (18 %) ^a	[91,92]
Heuristics and Metaheuristics (18 %) ^a	[90,94]
Column Generation (CG) (9 %) ^a	[88]

^a Percentages have been rounded to the closest number.

3.1.4. Computational modelling

Computational modeling including algorithms can be deployed in solving/calculating an answer to a given problem by establishing mathematical instructions/rules using appropriate software. Zhao et al. [68] developed an approach to evaluate wildfire evacuation by employing a large-scale GPS dataset that is input into a proxy-home-location interface algorithm, along with an evacuation

behavior interface algorithm (rule-based). Chen et al. [67] investigated the significant human factors that affect pre-evacuation time using surveys to construct an analytical framework based on the random forest algorithm. This algorithm is a machine learning method employed in applications requiring classification and regression. Multiple decision trees are combined in this ensemble learning technique for making predictions [83].

3.1.5. Choice models

Choice models are statistical models used to investigate and predict the decisions that a person or set of people make when encountering a variety of options. These models are often utilized to describe decision-making processes and predict how individuals decide among several options, which can be used in a wide range of areas, including economics, marketing, transportation, and social sciences [95]. Three approaches of Choice Models appeared in the literature; binary logit model [84], hybrid choice modeling [53], and latent class logit model [78], which were used to investigate exit choice, risk perception, and compliance respectively. The repeated binary logit model is recommended by many researchers for exploring interchanging decision-making during emergencies. It is typically used in evacuation research that focuses on predicting dynamic travel demand under various evacuation conditions. Thus, this technique yields accurate input data required to model traffic evacuation within road networks [17]. On the other hand, hybrid choice models have been developed as an improved version of discrete choice models to incorporate attitudinal variables. Bian et al. [78] investigated compliance with a phased hurricane evacuation order by employing a latent class logit model to analyze household behavioral intention survey. Using a discrete approximation, latent class logit models divide people into a predetermined number of classes based on their heterogeneity [96]. The modeling strategy differs from random parameter logit models, which describe individual heterogeneity using continuous distributions. The latent class logit model is thought to be less flexible than the random parameter logit model due to the discrete distribution. But one advantage of employing latent class models is that it eliminates the need to describe a distribution [97].

3.1.6. Multi-agent systems (MAS)

This technique is a prominent type of artificial intelligence and can solve complex problems that cannot be worked out with monolithic or single-agent systems. Tanaka et al. [47] employed a multi-agent simulation model in the context of the festival hall to study selfish behavior among evacuees. Keykhaei et al. [75] suggested a situation-aware evacuation framework that makes use of multi-agent modeling to study the behaviors, interactions between agents (vehicles, pedestrians, and management agents), and interactions between agents and the environment. Liu et al. [76] proposed the best way to construct emergency evacuation zones in areas of high density, in which the community's evacuation area and the evacuees' behavior patterns were investigated following the earthquake. Based on a multi-agent system, a simulation model for emergency evacuation in high-density communities was developed, taking into account changes in the community environment and the psychological and behavioral patterns of individuals.

3.1.7. Machine learning

Sun et al. [77] studied the evacuation decisions based on a survey dataset collected in the aftermath of Hurricanes Katrina and Rita by developing an interpretable machine learning approach, named enhanced logistic regression model. Anyidoho et al. [80] analyzed and predicted the choice of destination decisions by employing a machine learning approach based on survey data from four recent U.S. hurricanes.

3.1.8. Optimization approaches

An optimization problem is simply one of finding the extreme point that maximizes or minimizes the value of a given objective function of the variables which is itself a movable hyperplane [98]. Mathematical modeling/optimization approaches are not widely used in investigating human factors/behaviors. However, a study [42] developed an optimization model that sought to identify the locations of individuals who decided to Shelter-In-Place (SIP) by solving an assignment problem. Milburn et al. [81] investigated the nearest-shelter behavior by solving p-median and p-center optimization models with an objective function that reduces the total population-weighted distance between each demand center and its designated shelter in the context of a hurricane.

3.1.9. Field experiment

Given the practical challenges, a controlled experiment might not give results that match actual disasters but may provide indicative findings. One study [63] investigated human behaviors in a controlled-tunnel evacuation experiment to evaluate the impacting factors, and included video recordings, physical conditions (heart rate and blood pressure), post-experiment questionnaires, and interviews. This study showed that natural ventilation in the urban tunnel fire evacuation has beneficial effects on the evacuation process; evacuees have a tendency towards choosing their entrance as an escape exit; and individuals seek bright places. Further, ceiling openings impact the evacuation by increasing the visibility and sense of safety. Lastly, the authors found that light affected evacuation efficiency more than signs and fire alarms.

3.1.10. Cellular Automata (CA)

CA is a mathematical modeling technique that can be implemented in solving complex systems that involve a large number of simple homogenous entities with regional interactions. Unlike ABS, CA is a far less popular method, involving only one study [41], which simulated post-earthquake crowd evacuation from a room and an entire (educational) building.

3.1.11. Internet of Things (IoT)

This technology concerns exchanging data between devices, systems, or other technologies over communication networks. This technique can be employed in traffic management and evacuation planning, as it connects physical objects (e.g., evacuees and dri-

vers) attached to sensors, processors, or other technologies with other devices. A single study [69] used this method along with forecasting to estimate panic amongst evacuees, to identify the critical zones, and hence prioritize the evacuation orders.

3.1.12. Virtual reality (VR)

Virtual reality (VR) has become a more important tool in recent years for visualizing possible emergencies and for evaluating and planning human responses to them [99]. Virtual reality is "a completely synthetic experience where users are presented with only virtual contents" [99], page 2. VR-based training for disaster preparedness and response, as noted by Hsu et al. [100], provides challenges concerning budget, time, and safety to be resolved. Leon et al. [82] studied vertical-evacuation decisions by developing a user-focused virtual reality evacuation experiment to automatically gather 435 participants' choices for refuge to existing locations that may be used as tsunami vertical evacuation shelters. In addition, they employed questionnaires to gather data regarding the participants' knowledge of tsunami evacuation, their level of awareness of it, and the considerations that went into choosing their shelter. Finally, they looked at the location aspects of the chosen buildings using GIS. GIS-based analysis was created by post-processing the data that was automatically acquired from the VR experiences.

4. Summary

The review of human behavior prediction in large-scale disasters has used a wide variety of approaches to study different facets of human behavior. The most common method by far is ABS, although several other approaches exist; the predicted behaviors include, for example, the choice to evacuate, evacuation route choice/exit choice, walking speed, social behaviors (herding, competitiveness), compliance with evacuation orders, and travel mode choice.

4.1. Evacuation routing optimization methodologies

In this section, we focus on studies that have sought to perform evacuation routing optimization, and the methods used to achieve this. For example, studies that have employed mathematical programming, e.g., the Shortest-Path problem to find the optimal evacuation routing, were reviewed as well, since in the study of transportation and communication networks, shortest-path problems are by far the most fundamental and often occurring challenges [101]. Three main pillars of optimization studies can be in their objective functions, decision variables, and case studies to validate and test the constructed model. This section begins with a summary table, followed by specific methods used in path optimization from the reviewed literature.

As shown in Table 4, various methods can be applied to solve evacuation routing problems. Table 5 illustrates a summary of the solution methods used in the optimization models and the corresponding papers.

Methods were arranged (Table 4) in descending order according to their presence frequency in the literature. Traffic assignment and IP are the highest usage frequencies employed to optimize evacuation routing. Further details for each method are discussed in the following sections.

4.1.1. Traffic assignment

In 1952, Wardrop developed two traffic assignment principles [102]. The first principle states that the travel times between each Origin-Destination (OD) pair on all routes that are used are equal and less than the travel times that a single vehicle would encounter on any route that is not used. As a result, user equilibrium—an equilibrium traffic assignment pattern—is described by Wardrop's first principle. The second principle states that in a network, the total travel time is the lowest possible value. Thus, the system optimum—also known as the ideal traffic assignment pattern—is described by Wardrop's second principle.

In light of these principles, discussing the practical problems associated with applying these theories to real-world scenarios is imperative. The problem of traffic assignment was initially formulated in Ref. [103]. Traffic assignment involves assigning evacuees to evacuation paths to maximize the utilization of the traffic network and shelter capacity in the shortest possible time. Many evacuation planning and management models are derived from traffic assignment methods, such as user equilibrium, system optimal, nearest allocation, and constrained system optimal [21]. Different traffic assignment methods are used depending on the objective function. Nearest allocation assigns evacuees to their closest shelters, which may result in poor utilization of road capacity. A previous study focused on minimizing total evacuation time by creating a chronological evacuation plan that considers departure time and routing while avoiding congestion [85]. According to the authors, the developed model can be enhanced by addressing more issues like evacuees' behaviors. Another method of traffic assignment is the User Equilibrium, which was proposed in Ref. [104] to be modeled for a general network as a constrained nonlinear optimization problem. User equilibrium is defined as a traffic assignment approach that presumes intelligent evacuees who can identify optimal paths across a long period, depending on the concept of repeated games and/or the presence of comprehensive travel times guidance. Udentia et al. [86] provided a methodology for enhancing the effectiveness of a multi-modal transportation system in urban areas by evaluating chock nodes, influx, and affected zones in highway networks and the available preferences on evacuation routing. A further study [87] proposed a formulation that evaluates the functioning of a traffic system placed in the Central Business District (CBD) based on the concept of "connectivity".

4.1.2. Integer programming (IP)

Within the broad topic of mathematical programming, the most used modeling method intended to help managers in planning and decision-making is Linear Programming (LP) [105]. A further optimization approach that has been employed is IP, which uses integer variables in an LP model. IP problems are optimization problems in which variables must have integer values. Pure and mixed problem types are further classifications for IP problems. Pure IP problems are the type of problems where the solutions to all the decision variables have to be integers (general integer, binary, or both). Mixed IP problems are the type of problems where the solutions to

some of the decision variables—general integer, binary, or a combination of the two—must be integers, but not all of them. Fractional optimum values can be obtained for the noninteger variables [105]. A mixed variable problem arises when there are some continuous variables. An integer linear programming (ILP) problem is produced when all functions are linear; otherwise, it is nonlinear. It is possible to transform the ILP problem into a 0–1 programming problem. 0–1 programming problems can also be derived from linear problems involving discrete variables [106].

IP was employed in a study investigating the dynamics accompanied by gas diffusion evacuation to identify the securest paths for the evacuee depending on the numerical risk assessment [89]. IP has also been utilized to solve an optimal exit route assignment problem in the context of a large transportation network to move evacuees into safety [12]. Further, Kim et al. [93] integrated physics-based flood prediction models and stochastic optimization for large-scale multi-facility coordination of hospital and nursing home evacuations in response to an approaching hurricane. The optimization model addresses scenario-dependent evacuation demand, transport vehicles with different capacities, and both critical and non-critical patients.

4.1.3. Simulation-based approaches

Simulating different scenarios to imitate real-world events can be useful in evacuation planning since mimicking hazards in the real world is unsafe. Henry et al. [91] used a first-in-first-out queuing simulation to evaluate the clearance times with consideration of the evacuation demand by employing MATSim. A further study [92] investigated the self-evacuation organization by private vehicles.

4.1.4. Heuristics and metaheuristics

Metaheuristics are wide-ranging problem-solving models that direct the search for high-quality answers for sophisticated optimization problems, whereas heuristics are customized problem-solving strategies or rules of thumb used to identify approximate solutions rapidly [107]. Metaheuristics use heuristics as their foundation and create a systematic method for dealing with a broad range of optimization problems. Computational approaches such as algorithms play a significant role in both heuristics and metaheuristics optimization approaches.

Finding the shortest path or route from a starting point to a final destination is known as the "shortest path problem" [108]. Dijkstra's algorithm, which is an algorithm used to find the shortest path in an optimization problem, can be used in planning for evacuation routing. Dijkstra's algorithm employed in Ref. [90] proposed a compromised model that assesses the risk of the evacuation process in the context of toxic gas leakage. Han et al. [94] designed a method that employs the Dijkstra algorithm for evacuation route identification, the Random Forest algorithm for travel time estimations, and the influence of hazards and traffic congestion on evacuation time to evaluate evacuation vulnerability and risk based on evacuation time.

4.1.5. Column generation (CG)

In large-scale optimization problems, and in particular, where we have too many variables, column generation can be used. Here, variables are iteratively added to the model [109]. Column generation has been used to solve a hurricane evacuation optimization problem in a stochastic-dynamic network [88], which assumes that some input data values are not known with certainty.

5. Summary

The role of optimization in evacuation planning can be classified into two approaches: planning for the evacuation process and evaluating an existing evacuation plan [85]. Traffic modeling mainly concerns two important aspects: travel demand and traffic assignment and can assist in evacuation planning and real-time evacuation [26]. Optimization approaches have a significant impact on planning for evacuation routing by improving the process of assigning the most efficient paths for evacuating residents from areas prone to hazards. Different techniques can be used which vary depending on a few factors like the objective of the optimization model and hazard type and scale. Analyzing optimization models' results enhances evacuation plans' effectiveness, safety, and efficiency, assisting in ensuring uninterrupted and timely evacuation during emergencies.

5.1. Interdisciplinary approaches

Table 6 shows the evacuation scale/type, optimization approach and model objective, human behavior prediction method, and specific human factors investigated in previous research which has integrated human factors with optimization approaches. That is, while all papers reviewed within sections 3.1 and 3.2 referred to both human factors/behavior and optimization with a focus on one or the other, the papers in Table 6 adopted approaches that integrated both.

As shown in Table 6, various methods can be applied to combine human factors/behavior investigation into the optimization/simulation of evacuation routing models. Table 7 illustrates a summary of the interdisciplinary methods and the corresponding papers.

Table 7 shows that the most used technique in incorporating human factors/behavior into evacuation route planning is simulation platforms/software. Further details are discussed in the following.

5.1.1. Heuristics and metaheuristics combined with human factors/behavior studies

Employing algorithms in solving optimization models is a widely used technique in planning evacuation routes, which can be combined with several human factors/behavior modeling techniques, such as ACO combined with CA [110], Dijkstra's algorithm along with Type-II fuzzy logic system and logit model [111], Dijkstra's algorithm along with attraction angle model [112], Dijkstra's algorithm along with SDT [114], Dijkstra's algorithm with ABS [124], and A* search algorithm for finding the evacuation routing and Fuzzy Inference System (FIS) and Deep Long Short-Term Memory (DLSTM) algorithm to model evacuees' situational awareness [128]. In a study by Wang et al. [129], the K-means algorithm and an enhanced parameter adaptive DBSCAN algorithm were used to

Table 6

Interdisciplinary methodologies presented in the literature.

Ref.	Hazard/Evacuation/scale	Optimization Aspects		Human Factors/Behaviors Aspects	
		Method	Model Objective	Method	Factors/Behaviors
[110]	Urban development land/stadium	ACO	Find the shortest path based on the herd behavior and the trend of the Shortest Path First	CA	Seeking the shortest path first, herding behavior, and the relationships among evacuees
[111]	Large-scale	Dynamic network loading, Dijkstra's algorithm & Monte Carlo (EVAQ)	Maximize the total number of evacuees arriving at the secure areas	Type-II fuzzy logic system/logit model/TIL	Route choice behavior (panicky and rationale individuals)
[112]	Evacuation risk assessment	Dijkstra's algorithm	Calculate the shortest path by solving the classic shortest path problem (O-D assignment problem)	Attraction angle model	Route choice (rationality in risk perception during evacuation)
[113]	Individual/network of smart buildings	Traffic assignment (system optimal)	Calculate the shortest, and hence, safest routes	Traffic assignment (system optimal)	Herding behavior (stress and frustration) and route familiarity (route desire rate)
[114]	Urban/flood	Dijkstra's shortest path algorithm within a GIS environment using Avenue® programming	Minimize total disutility (distance, time, and general physical link)	SDT	Evacuees' perception of flooding (river buffer and elevation)
[115]	No-notice/large-scale	Dynamic traffic assignment/VISUM built-in assignment method/Dynamic User Equilibrium	Identify the ideal evacuation routing	Decision Tree (classification and regression tree) using R	Mode choice
[116]	Post-disaster	Mixed-integer linear programming	Minimization of panic-generated psychological penalty cost, psychological interference cost, and costs related to transportation and construction shelters, wherein they aggregated into a single objective by assigning weights	Binary logit model using TRANSIMS The Susceptible–infective–removal (SIR) model	Gathering behavior Panic spread
[117]	Large-scale/dam-break/flood	Microsimulation (Legion)	Modeling large-scale evacuation scenarios	Statistics (Normal distribution applied to previous data) Survey	Pre-evacuation (recognition and response) time Age, gender, ability to use the stairs, knowledge (evacuation policies, assembly points, etc), warnings response, routing choice, helping behavior, use of vehicle, and seeking information
[118]	Wildland short-/no-notice	Traffic assignment – system optimal using Python	Optimization of evacuation instructions (departure time, routing, and safe zones) by minimizing total travel cost	Sampling Average Approximation (SAA)/routing variabilities (main = compliance and secondary = non-compliance)	Compliance with evacuation instructions (departure time, routing, and safe zones)
[119]	Wildland urban	WUI-NITY platform	Identify the vulnerability of an area over time	WUI-NITY platform	The departure time is influenced by the decision to evacuate, enter the traffic node, and comply with evacuation orders to identify the number of evacuees entering the transportation network

(continued on next page)

Table 6 (continued)

Ref.	Hazard/Evacuation/scale	Optimization Aspects		Human Factors/Behaviors Aspects	
		Method	Model Objective	Method	Factors/Behaviors
[120]	Hurricane/coastal areas	TRANSIMS platform	Optimization of the picking-up sites, scheduling, and routing for bus evacuation	ABS	Demographics (age, impairment, job condition, employer, travel mode, and distance to workplace), household type, family size, the decision to evacuate influenced by herding behavior and instructions compliance
[121]	Wildfire	Quasi-dynamic traffic assignment using TransCAD/Monte Carlo	Network reliability & vulnerability assessment	Tridmatrix/NYBPM Second Generation	Time-dependent travel demand influenced by auto-ownership, family size, and evacuation decisions
[122]	Earthquake/crowd	O-D cost analysis/multi-agent systems/GIS	Identify the optimal evacuation routing (given exit familiarity)	CA/continuum model	Exit choice considering exit familiarity, visibility, and panic effects
[123]	Tsunami	REVASIM	Minimize total travel time for all agents by solving linear programming optimization problem	ABS	Compliance and herding behavior affecting (evacuation decision, mode-route choice, destination choice, population movement, and shelter arrival)
[124]	Urban/flood	Dijkstra's algorithm	Find the shortest path between nodes	ABS using NetLogo/Monte Carlo	Evacuation decision, speed, evacuee's vulnerability, risk perception, evacuation route search, risk acceptance threshold, and mutual aid process
[125]	Flood	Dynamic traffic assignment	Traffic simulation	ABS	Self-evacuation activities
[126]	Wildland-urban interface fires	WUI-NITY platform	Traffic simulation with the incorporation of different smoke intensity levels	WUI-NITY platform/Unity 3d game engine	Driving speed influenced by smoke presence
[127]	Mass/wildfire	WUI-NITY platform	GIS-based traffic, pedestrian and routing models	Stochastic flow-based approach using the WUI-NITY platform	The decision to evacuate and the movement towards vehicles
[128]	Earthquake/crowd	A* search algorithm/agent-based simulation/NetLogo	Provide a situation-aware system to reduce the risks, guarantee the effectiveness of the evacuation plans, promote and estimate both current and potential operations within the dynamic space of the crisis	FIS/DLSTM/ABS/NetLogo	Individuals' awareness of their surroundings and other agents' situations, as well as their understanding and sense of environment (influenced by age, gender, health level, weight, speed, congestion rate, distance from congestion, and elapsed time)
[129]	Mass/pedestrian	ACO/CA	Decrease the average completion time, avoid congestion, optimize network usage, and effectively distribute exit loads	CA	A variety of pedestrian behaviors and how they interact with the environment. Congestion avoidance
[130]	Tsunami/tsunamigenic earthquake	Dijkstra's algorithm/ABS	Evaluate the impact of the micro-scale urban vulnerabilities	ABS/collision avoidance algorithm	Starting time of the evacuation, pedestrian flow and speed, evacuation delay
[131]	High-risk emergencies (accidents at chemical and fire-explosion hazardous facilities)	Fuzzy logic/GIS/Petri.Net Simulator	Developing theoretical and methodological basis that integrate cognitive and dynamic models to address real-world issues pertaining to emergency mitigation and prevention	Fuzzy logic/machine learning	Evacuee's status (identified by speed, position, and the relative route of the main group) and their interaction with the surroundings and other evacuees

(continued on next page)

Table 6 (continued)

Ref.	Hazard/Evacuation/scale	Optimization Aspects		Human Factors/Behaviors Aspects	
		Method	Model Objective	Method	Factors/Behaviors
[132]	Self-/flood evacuation	Traffic simulation/RoadLoad application/GIS OpenStreetMap/Topographic Objects	Identify the local determinants for self-evacuation	Questionnaire survey	Knowledge of evacuation procedures, mode choice, ability to self-evacuate, vulnerable individuals, awareness and preparedness, and demographics and socioeconomics
[133]	Urban/hurricane	RL/Dijkstra's algorithm/OpenStreetMap	Provide an effective way to improve the effectiveness, safety, and reliability of emergency operations in dynamic urban areas	RL	Evacuee's state (influenced by current position and destination choice)

ACO- Ant Colony Optimization, TIL- Travel Information Level, SDT- Signal Detection Theory, ABS- Agent-Based Simulation, GIS- Geographic Information Systems, RL- Reinforcement Learning, DLSTM- Deep Long Short-Term Memory, FIS- Fuzzy Inference System.

Table 7

Methods used in solving evacuation routing problems in the literature.

Method	Research
Heuristics and Metaheuristics combined with human factors/behavior studies (33 %) ^a	[110–112,114,124,128–130,133]
Traffic Simulation Platforms/software combined with human factors/behavior studies (29 %) ^a	[117,119,120,123,126,127,132]
Traffic Assignment combined with human factors/behavior studies (21 %) ^a	[113,115,118,121,125]
Mixed-integer programming (MIP) combined with human factors/behavior studies (5 %) ^a	[116]
Multi-agent systems (MAS) combined with human factors/behavior studies (5 %) ^a	[122]
Fuzzy logic combined with human factors/behavior studies (5 %) ^a	[131]

^a Percentages have been rounded to the closest number.

organize pedestrian evacuation and classify pedestrians into appropriate groups. Furthermore, the enhanced CA model determines the evacuation path for pedestrians who are free to evacuate. The improved ACO algorithm generates the evacuation routes for each organized evacuation group, and it schedules them according to the evacuation aims and collision avoidance technique. Another study conducted by Cienfuegos et al. [130] employed Dijkstra's algorithm and used micro-scale ABS to quantify the delay in evacuation procedures caused by the existence of urban micro-vulnerabilities. An RL-based routing algorithm was proposed by Li et al. [133] to support individuals in navigating around urban areas during flooding disasters under complicated, information-limited, and dynamic road network scenarios. To reduce loss and enhance the long-term gain of an intelligent agent, deep reinforcement learning (DRL), a type of machine learning technique, is currently being utilized in such complex sequential decision-making processes [134].

5.1.2. Traffic simulation platforms/software combined with human factors/behavior studies

Human factors/behavior were investigated using a survey and statistics, then input into a microsimulation software, named Legion, to model large-scale flood evacuation scenarios [117]. Online platforms such as WUI-NITY can be used to efficiently plan for evacuation routes with the incorporation of human factors/behavior in the context of wildland-urban interface fires [119,126,127]. Additional simulation platforms like TRANSIMS which can solve optimization models while accounting for the findings from human factors/behavior investigation using ABS is an effective approach to attain more realistic evacuation planning strategies [120]. REVASIM has also been used with ABS to solve an optimization problem along with including the dynamic of evacuees' behaviors (e.g., compliance and herding behavior) [123]. Borowska-Stefanska et al. [132] employed traffic simulation software along with a questionnaire survey and evaluated the local factors that influence self-evacuation in the context of flooding and the effect that alterations to its different phases and forms have on the regional transportation system's operational effectiveness.

5.1.3. Traffic assignment combined with human factors/behavior studies

Similar to the optimization literature presented in section 3.2, a significant number of interdisciplinary papers used this approach only [113] or along with human factors/behavior investigation approaches, which are decision tree and binary logit model [115], Sampling Average Approximation (SAA) [118], Tridmatrix/NYBPM [121], and ABS [125].

5.1.4. Mixed-integer programming (MIP) combined with human factors/behavior studies

Utilizing the MIP in incorporating human factors/behavior into evacuation route planning is not a commonly used technique, which was used only once with the Susceptible–infective–removal (SIR) model [116].

5.1.5. Multi-agent systems (MAS) combined with human factors/behavior studies

As mentioned earlier in section 3.1.9., MAS is capable of solving complex models. The results from a CA study input into MAS to find the optimal evacuation routing with consideration to exit choice preferences and panic effects [122].

5.1.6. Fuzzy logic combined with human factors/behavior studies

Fuzzy set theory can be utilized in decision models to incorporate unquantifiable, inadequate, and unobtainable information as well as facts that have been partially ignored [135]. Fuzzy logic and fuzzy set theory were first presented by Zadeh [136] and are used to explain ambiguity and uncertainty in decision-making. "If the input variable is a fuzzy set, THEN the output variable is a fuzzy set" format is the linguistic rule that defines fuzzy logic. To function, fuzzy inference systems will first involve "fuzzification"; in this step, input data that is crisp at first is converted to fuzzy data that indicates the extent of a person's membership in preset fuzzy groupings; later, to get a clear output from a fuzzy system, defuzzification is a required step [135]. Danilchenko et al. [131] employed membership functions and fuzzy sets to reflect and capture the uncertainty and inconsistency that are intrinsic to the evacuation process.

6. Summary

Several approaches for integrating human factors/behavior into evacuation routing optimization/simulation are presented in the interdisciplinary literature. For instance, human factors/behavior data obtained from CA, fuzzy logic systems, decision trees, binary logit models, ABS, machine learning, and questionnaires can be further input into evacuation routing optimization/simulation.

7. Human factors/behaviors investigated in the reviewed papers

Based on the reviewed literature that investigates human factors/behaviors, the most common and significant human factors/behaviors considered in all of the review literature are listed in Table 8.

Evacuation decisions represented the most investigated human behavior in the human factors/behavior literature. **The decision on whether to evacuate or not** is one of the most influential factors, as it directly affects evacuation demand, which can significantly influence traffic flow, congestion rate in real-time evacuation, and planning for roads' capacities within transportation networks [119,121].

Route choice is another crucial subject that should be considered when planning for evacuation [24]. For instance, assigning evacuation routes that lead to higher compliance rates is an effective strategy for minimizing the negative impacts of a disaster and achieving efficient evacuation [118]. Bakhshian and Martinez-Pastor [33] found that route-choice behavior is influenced by factors such as gender and age. Wu et al. [137], however, demonstrated that certain demographics—such as gender, age, income, and ethnicity—are not reliable indicators for predicting evacuation logistical variables (i.e., aspects and variables that affect how well evacuation procedures are planned, carried out, and maintained). Notably, route decisions have several influences that can affect them dramatically, such as obstacle avoidance behavior, herding behavior, panic, familiarity, and compliance with evacuation orders. Therefore, route decisions should be investigated simultaneously with these associated factors.

Social interactions, such as grouping behavior and collecting family members, can affect departure time, routing choice, and destination choice [44]. Treating the evacuees as a homogeneous group can result in reducing the validity of the simulation studies. Although eliminating social interaction issues from evacuation planning makes it less complex and provides more optimistic evacuation times [38].

Destination choice can vary from one individual to another; for example, individuals who follow evacuation instructions (in case of availability and there is no information flow interruption) will seek assembly points or designated shelters [87]. **People who have relatives, friends, and children to be collected will head directly to the intended person** [24]. Some will go to the highest areas in the case of tsunamis [56], while others will escape the town far away from the affected zone. Examining the variabilities in destination choice is of great value in designing for shelter's capacities and locations.

Table 8

The most commonly investigated human factors/behaviors in the literature.

Human Factors/Behaviors	Research (69 papers)
Evacuation Decision (30 %) ^a	[38–41,51–53,56,57,59,68,73,74,77,78,119–121,123,124,127]
Route Choice (22 %) ^a	[41,46,54,61,63,67,71,75,111,112,117,118,123,124,128]
Social Interaction (17 %) ^a	[38,44,51,54,58,64–66,72,75,110,131]
Evacuation Movement and Speed (17 %) ^a	[42,60,64,71,72,75,79,128–131,133]
Destination Choice (14 %) ^a	[51,66,73,78,80–82,118,123,133]
Vulnerability (14 %) ^a	[39,41,42,47,55,61,79,117,124,132]
Departure Time (13 %) ^a	[40,53,63,68,71,78,118,119,130]
Demographics (13 %) ^a	[38,39,51,67,75,120,121,128,132]
Risk Perception (13 %) ^a	[38,50,51,53,59,78,112,114,124]
Mode Choice (12 %) ^a	[40,49,78,115,117,120,123,132]
Panic Behavior (10 %) ^a	[45,49,69,76,111,116,122]
Herding Behavior (9 %) ^a	[46,49,110,113,120,123]
Familiarity (9 %) ^a	[50,54,61,75,113,122]
Compliance Rate (7 %) ^a	[68,78,118,120,123]
Exit choice (6 %) ^a	[43,46,58,122]
Situational awareness (3 %) ^a	[75,128]
Cooperative behavior (1 %) ^a	[76]

^a Ratios have been rounded to the closest number.

Risk perception is one of the greatest factors impacting the evacuation process. It can affect departure times, evacuation decisions, and other aspects [53]. Evidence suggests that risk perception is the factor with the most extensive impact on evacuation decision-making [138]. People who do not feel the urgency to leave will shelter in place; on the contrary, evacuees who notice cues will sense the danger and act upon it [59]. Further, studies indicate that risk perception is highly correlated with socioeconomic characteristics, including age, household size, professional status, home ownership, number of working family members, monthly income, and income source [139]. On the contrary, Lovreglio et al. [53] showed that there is no relationship between age, gender, or wealth and risk perception.

Incorporating **herding behavior** in planning for emergencies can reflect more realism in response strategies, as people may be influenced by others' decisions in real life. Seeing relatives, friends, and neighbors, or even groups of strangers, evacuating, or selecting routes/targeted destinations could encourage some of the evacuees to follow their leads [49]. Consequently, thinking about the potential outcomes of such behavioral conduct can allow government authorities to take corrective measures in implementing their tactics. A study evaluated the self-organization effect on the evacuation process and provided a supporting conclusion [122].

Familiarity is mainly concerned with **routing and destination decisions**. Accordingly, it will have a direct effect on the **congestion** intensity of the road network, as many residents will choose to evacuate through familiar routes instead of following evacuation orders [24,118].

Several human factors/behaviors were only investigated in a limited number of papers, like emotional state [39,62], competitiveness [46–48], grouping behavior [49,61,115], stress [44,50], obstacle avoidance behavior [54], pre-evacuation time [67], evacuees' movements toward vehicles [127], visibility [122], evacuation behavior of older people [70], driving speed influenced by visibility [126], evacuees' locations [42], and warning response [117].

8. Optimization models parameters in the reviewed papers

Traffic assignment models serve as the foundation for the majority of the current evacuation planning and management methods [21]. Interestingly, this fact agrees with this work's findings since the greater portion of evacuation routing methodologies in the reviewed papers comprised traffic assignment approaches.

The ultimate objective of the papers reported, as seen from a system perspective, is to **reduce the overall evacuation or network clearance time** [140]. In this review, traffic assignment was used with various methods and objective functions to achieve this [85–87,115,118,121,141]. These include, for instance, the nearest allocation technique [85] and user equilibrium with its different states (static and dynamic) [32,86,87]. A quasi-dynamic traffic assignment appeared in the literature [121], which can be implemented in an online traffic platform. The traffic assignment system optimal approach was deployed by two studies [113,118]. The objective functions that were attained with traffic assignment methods vary between minimizing the total evacuation time [85,86], maximizing traffic network efficiency [87], optimization of evacuation routing concerning distance and/or safety [113,115], minimizing total travel cost [118], and reliability and vulnerability assessment [121].

The second approach that was commonly used in the optimization of evacuation routing is the employment of algorithms, including ACO, Dijkstra's algorithm, and minimum-cost maximum-flow algorithm, with **Dijkstra's algorithm being the most frequently used technique**. This is attributed to its main objective of identifying the shortest path, given that **the main goal of most evacuation procedures is evacuating as many people as possible in the least time**, which can be attained by assigning evacuees to the shortest paths. To solve the routing problem, Dijkstra's algorithm was used in studies by Chen et al. [111], Chen et al. [112], Lee et al. [114], and Wang et al. [124]. Ant colony optimization algorithm was used in a single study that aimed to incorporate both human factors and routing, as this approach specifies a driving force (human factor) to identify the routing direction [110]. Jiang et al. [90] combined Dijkstra's and minimum-cost maximum-flow algorithms. The objective function incorporated in the problem solved using algorithms involves optimization of evacuation routing with respect to distance, risk assessment, driving factors, and time [90,110,114,124] and maximizing the total number of evacuees arriving in secured areas [111].

The literature reports many simulation techniques that can be utilized in traffic and transportation investigations, including online platforms, advanced simulating software, and queuing theory. A well-known method used for measuring traffic features (e.g. vehicle lines, road capacity, waiting times) is queuing simulation. FIFO queuing simulation through MATSim software was used to measure the variation in clearance times [91]. Traffic simulation is also used in investigating self-evacuation organizations by private vehicles [92]. However, many online platforms and computer software are valuable in transportation management and can also be applied in traffic planning during emergencies. A large-scale evacuation scenario was simulated using microsimulation software known as Legion [117]. Online platforms, such as WUI-NITY and TRANSIMS, were mentioned in the literature; the former was exploited to implement a vulnerability assessment of a predetermined area [119], whereas the latter wielded in a multimodal evacuation (specifically, bus) to provide optimal decisions regarding pickup locations, scheduling, and routing [120].

Methods such as **column generation, integer/mixed-integer programming, and linear programming** were also utilized to find the **optimal evacuation routes**. One study used column generation together with EDTA intending to minimize the total flow time or makespan within the transportation network [88]. Other researchers have utilized integer/mixed-integer programming to measure different parameters, such as prospective risk and transportation cost, and hence optimize evacuation routing [89,116]. Lastly, an assignment optimization problem was solved using LP to minimize the total travel time of evacuees [123]. Hybrid approaches like Multi-agent systems (MAS) were reported in the optimization of exit familiarity among evacuees alongside the employing of O-D cost analysis and GIS [122].

9. Hazard contexts

Table 9 illustrates the hazard contexts (if available) in the reviewed papers.

Evacuation planning research shows that large-scale evacuation in the context of flood or hurricane is the most frequently investigated approach. Followed by tsunami, earthquake and chemical leakage, wildfire, and volcanic and terrorist attacks. Nuclear incidents and human-made fires are minimally covered, appearing only once each. Evacuation routing optimization techniques are notably absent in wildfire, earthquake, volcanic, and human-made fire contexts. Interdisciplinary studies overlook all types of human-made disasters and volcanic emergencies. In human factors literature, all disaster contexts are represented, with limited studies on chemical leakage, terrorist attacks, and nuclear incidents. While fire evacuation is mentioned only once in human behavior studies, it is not overlooked; rather, it is excluded due to a focus on building evacuation contexts.

10. Discussion

This review presents methods that enable the researcher to model and evaluate human factors/behavior, as well as plan the evacuation routing under hazardous situations. To help researchers apply the most appropriate technique for their problem, it is still important to classify methods based on the nature of the problem (e.g., hazard context and investigated factors/behaviors). The majority of prior research was not focused on the approaches and methods used for studying human factors/behavior and/or routing optimization throughout the evacuation process. Thus, the initial aim of this review was to highlight and evaluate the proposed approaches in the evacuation planning literature.

In predicting/modeling human factors/behavior during emergencies, the ABS approach was the most frequently used technique in the prior literature with a percentage of 43 % (in the human factors/behavior literature). Wafda et al. [142] emphasized that the ABS approach is widely utilized in the development of emergency evacuation simulations owing to its agent-like ability to evolve according to the surrounding environment. Further, Lee et al. [38] reviewed prior academic and scientific work and reported that a significant amount of research employed ABS to simulate human behavior during emergencies. These conclusions are consistent with the systematic review findings, which indicated that a vast majority of the studies employed ABS to investigate human factors/behavior in the context of large-scale disasters.

In planning for evacuation routing literature, traffic assignment, and IP are the most employed methods with an equal percentage of 27 %. This evidence supports the assertion made by Bayram [21] that traffic assignment models serve as the foundation for the majority of the current evacuation planning and management methods.

In interdisciplinary research, heuristics and metaheuristics approaches are the commonly used methods in planning evacuation routing, while accounting for some human factors/behavior with a percentage of 33 %. Such conclusions are confirmed in Dulebenets et al. [143]'s research, which suggests that numerous heuristic-based algorithms were employed in an extensive amount of research to solve emergency evacuation problems. Metaheuristic techniques have also been employed alongside heuristics to identify high-quality solutions to emergency evacuation planning problems.

Further, the decision to evacuate was the most frequently investigated human factor/behavior with a percentage of 30 %. According to Lim et al. [144], evacuation planning and modeling have involved an intensive examination of the factors influencing an individual's or household's decision to evacuate. This finding aligned with the review's conclusion, in which a great portion of the reviewed papers investigated the decision to evacuate. Evacuation demand estimation during an emergency requires modeling evacuation decisions since these decisions entail complex behavioral aspects that take into account social and environmental considerations. Authorities must first understand the conditions around the people or households who decide to evacuate [144].

The majority of the conducted studies assumed either a flood or hurricane evacuation context with an equal percentage of 21 %. As Khalili et al. [145] observed, hurricanes and floods are relevant in the context of transit-based evacuation planning, considering their catastrophic effects on urban regions and the need for effective evacuation precautions. Similar to this, Yin et al.'s evaluation [146] of strategic evacuation planning for coastal areas noted that floods—especially those caused by storms—are a major concern because of the extensive damage they bring to people and infrastructure.

Table 9
Hazards' contexts in the reviewed papers.

Context	Studies (58 papers)		
	Human Factors Literature	Optimization Literature	Interdisciplinary Literature
Flood (21 %) ^a	[42,62,64,70]	[85,92,94]	[114,117,124,125,132]
Hurricane (21 %) ^a	[38,55,59,66,77,78,80,81]	[88,93]	[120,133]
Tsunami (16 %) ^a	[40,52,56,71,72,82]	[91]	[123,130]
Chemical leakage (12 %) ^a	[50]	[89,90]	–
Earthquake (12 %) ^a	[39,41,75,76]	–	[122,128,130]
Wildfire (9 %) ^a	[53,68]	–	[121,126,127]
Volcanic (3 %) ^a	[51,57]	–	–
Terrorist attack (3 %) ^a	[60]	[87]	–
Nuclear incident (2 %) ^a	[49]	–	–
Fire (2 %) ^a	[63]	–	–

^a Percentages have been rounded to the closest number.

The following sections illustrate limitations and future suggestions of methodologies employed to model/predict human factors/behavior, along with planning evacuation routing in large-scale emergencies.

10.1. Human factors/behavior aspects in evacuation planning

10.1.1. Methodologies

Based on the reviewed papers, investigating human factors/behavior can be implemented using several approaches that vary according to the purpose of the research and the problem nature. Table 10 highlights the important aspects of frequently used methods in analyzing human factors/behavior in hazardous situations, which are ABS, SFM, CA, and logit models, identifying some pros and cons of the proposed methods.

Table 10 illustrates some similarities and differences between the methods used in human factors/behavior investigations. For instance, ABS, CA, and SFM can be used in studying human factors/behavior at microscopic scale, which refers to individually identified entities [150]. On the contrary, ABS assumes a heterogenous flow of the crowd, unlike CA which assumes a homogeneous flow of the crowd. According to Bakhshian and Martinez-Pastor [33], while agents are a part of both the CA and ABM methods, the agents in CA are static and only communicate with their neighbors, whereas the agents in ABM are dynamic and their neighbors can vary based on the time of day. To model the formation of fixed objects and characteristics, like the establishment of the community's populations, CA is perfect. However, the ABM is the recommended method for assessing individuals' behaviors and interactions over time. Instead of limiting researchers to a basic system like CA, ABM offers them a system that is nearly comparable to reality. Rather than replicating an infinite grid, CA is commonly employed to simulate situations with a finite grid. Additional significant drawbacks of CA include its propensity to update the state with neighboring states based on simple rules and its extremely long run duration.

The logit models are a suitable method for formulating discrete choice problems. Employing logit models has the benefit of being remarkably simple to apply in linear models (unsuitable for nonlinear models). Besides, a transport network's route choice behavior can be predicted with the help of logistic base models. Nonetheless, the requirement that there be an association between the variables and the data is the primary drawback of these models; otherwise, the model's estimate could become less accurate [33].

Table 10
Recommended methods for investigating human factors/behavior during emergencies [39,147–149].

Criteria	ABS	CA	SFM	Logit Models
Evacuation Scale	Microscopic	Microscopic	Microscopic	Modeling probabilities
Crowd representation	Autonomous agents	Single element	Individuals subject to social forces	Sigmoid function/independent variable
Crowd behaviour	Heterogeneous/free decision-making based on the surroundings and the assigned rules	Homogenous flow	Identified by a driving force to reach a certain destination while maintaining a specific distance between the surroundings	Evacuation utility influenced by specified conditions
Realistic degree	High	Moderate	High	High
Usage	Preferred in behavioral studies including complex behaviors of agents and their surroundings (environment & others)	Favored in grid simulation spaces or transition probabilities are identified and constant	Describing the self-organization of several monitored collective impacts of pedestrian behavior	Modeling the dynamic travel demand based on utility function given certain conditions such as risk perception
Appropriateness with behaviors	Evacuation decision, route choice, herding, obstacle avoidance, competitiveness	Influence of exit width and obstacles on crowds, model friction forces and congestion, herding behavior	Exit choice, social interaction, evacuation movement, competitive behavior	Socio-demographics, compliance, familiarity, and herding behavior
Software	Python, NetLogo, Repast, MASON	Egress, Exodus, Steps	Maces, Hidac	SPSS, R, SAS, Python, Stata, NCSS, Java, MATLAB
Pros	Capable of assigning unique behavior for the agents, induces a natural description of the system, and flexible	High efficiency, robust scalability, and easy to use	Ability to describe different phenomena like road generation, congestion, and torque effects at exits	Prediction purposes (binary or multi-category levels), easy to implement and interpret, can be updated easily and reflect the importance of the predicted variables
Cons	Need more extensive computations than other models	Need extensive computations, not capable of considering high-stress behavior	Need extensive computations	Can't solve dynamic (non-linear) problems, over-fit with intensive datasets, and can't capture complex correlations
Alternatives	Game theory, fluid dynamics model, CA, and SFM	Game theory, fluid dynamics model, ABS, and SFM	Game theory, fluid dynamics model, ABS, and CA	Probit Model
Output Nature	Descriptive results driven by evidence	Transform non-linear correlations among different components of the system into feasible programs	Equations of motion are nonlinearly coupled Langevin equations	Probability distribution/consistent covariance matrix

Other methods that were not mentioned in Table 10, for example, VR, can be very useful in studying human factors/behavior in emergencies. There would be a major ethical dilemma if an evacuation experiment were carried out in the real world and peoples' safety or health were placed at risk. Fujimi and Fujimura [151] suggested employing VR technology for experiments to tackle these issues. Additionally, a virtual reality experiment gives researchers the ability to manipulate a variety of experimental variables, particularly those that are not achievable in the real world. This technique may virtualize an emergency and place several artificial individuals in it. It can thus predict behavior in a way that is fairly accurate and serve as an educational tool for humanitarian operations [33].

Having discussed the utilized methods in the human factors/behavior literature, moving on now to consider approaches that appeared in the interdisciplinary research. Although there are several papers incorporate human factors/behavior findings into the planning of evacuation routing, it is still a rich area that needs to be advanced further, considering the robustness and realism it brings to the outcomes of evacuation plans. This research indicated that ABS is amongst the most used techniques in studying human factors/behavior, hence, translating its outputs into evacuation route planning. WUI-NITY platform and CA are also commonly used methods to get numerical findings from behavioral studies to plan for evacuation routes efficiently. However, few researchers considered utilizing methods such as fuzzy logic, SDT, and decision trees in the interdisciplinary research.

To conclude, there is no particular method that can fit in all human factors/behavior studies, instead, different contexts identify the appropriateness of employing the suitable method. This research encourages the employment of methods that can stand alone and have wide commercial recognition, i.e., ABS. Moreover, advanced technologies like VR, augmented reality, and mixed reality that give highly realistic outcomes could be of great use yet did not feature strongly in the reviewed papers.

10.1.2. Human factors/behavior

Some other human factors/behaviors were not identified via the review process, but they are important since they impact the evacuation outcomes. For example, in a coastal New England state, West and Orr [152] showed that perspectives regarding sources of information, race, gender, and spatial proximity influence people's perceptions of natural disasters. Moreover, evacuation intentions are influenced by media and government reports encouraging people to leave, presence of children, and living near water [152].

Lee et al. [153] revealed that a few significant demographic and socioeconomic factors, including age, household income, and type of residence, played an essential part in the decision to evacuate. Age in particular influences an individual's tendency to leave. When disaster hits, it has been demonstrated that those under 30 are less likely to leave than individuals in other age groups. Seniors are less likely to leave their homes, according to Cutter and Barnes [154]. With regard to the reviewed paper, the evacuation behavior (walking/mobility) of older people was examined in a study conducted by Nakanishi et al. [70]. Further, it has been revealed that the size of the household has an impact on the decision to evacuate. During emergencies, people are more likely to leave homes with more than three members [153]. According to Lee et al. [153], household income has a significant influence on the decision to evacuate. People are more likely to evacuate when they have a higher household income; they are less likely to do so with a lower monthly income. Additionally, it is discovered that those who reside in multi-family housing evacuate at a higher rate than those who do so in detached housing.

The effects of natural disasters are more significant for elderly people, ethnic minority groups, people living in rural areas, and individuals with disabilities, affecting these groups in a disproportionate and negative manner [155]. In a study conducted by Dostal [156] shows that a large number of elderly persons who remain bound to their houses are unable to leave their neighborhoods and homes because of their physical disabilities and insufficient resources. This study also brought attention to the requirements of elderly people who are home-bound for shelter. Homebound elderly individuals' chronic healthcare needs were similar to those of nursing facility patients, emphasizing the complicated resources needed to keep this study's population securely accommodated in special needs shelters while preventing an unimportant load on healthcare systems.

Exit visibility has not appeared much in the literature, as the scope of this work was large-scale incidents, and most likely that exit visibility is more associated with building evacuation. Nonetheless, this factor is substantial with regard to routing decisions and could be applied further in network flow-based evacuation research.

In summary, in the context of large-scale evacuation planning, this research suggests the investigation of a manageable number of human factors/behaviors that may influence traffic network circumstances. For example, examining the decision to evacuate while accounting for compliance with evacuation instructions (i.e., departure time), along with examining route familiarity behavior in the traffic network. The former will provide a proper understanding of the dynamic of the evacuation demand. Whereas the latter could give useful insights into places where congestion might arise.

10.2. Evacuation routing optimization

Numerous approaches are available for optimization, such as evolutionary algorithms, heuristics, metaheuristics, interior-point methods, trust region methods, convex optimization, linear programming, quadratic programming, and conjugate gradient methods [157]. Consequently, there is a wide range of exact and heuristic methodologies available for solving optimization problems. Furthermore, the rise of artificial intelligence brought in an entirely novel generation of intelligent optimization techniques that can easily provide near-optimal solutions to challenging real-world optimization problems. These kinds of problems would not have been feasible to solve with traditional or exact optimization methods [107].

In the planning for the evacuation routing literature, several methods have been used to achieve the models' objectives. These methods include traffic assignment, IP, simulation-based approaches, shortest-path algorithms, and column generation. This review indicates that traffic assignment and IP are the most frequently used compared to other techniques. According to Bayram [21], the majority of evacuation models are derived from traffic assignment models. Traffic assignment includes assigning evacuees to evacuation paths based on user equilibrium or system optimal principles. The interdisciplinary research showed that the traffic assignment

approach can capture some human factors/behaviors like herding behavior, mode choice, compliance, evacuation decision, self-evacuation behavior, and demographics. Unlike traffic assignment, IP was able only to capture the panic spread in shelters. Although dynamic traffic assignment models provide a more accurate representation of traffic flow over time, they are not feasible when the size of the evacuation transportation system increases and have difficulty addressing large instances precisely. Furthermore, traffic assignment models require more computational work to solve particularly when networks and O-D pair counts are high [21].

Unlike traffic assignment that was employed in several interdisciplinary research, IP was able only to capture the panic spread in shelters. IP does not allow for detailed sensitivity analysis and is much more difficult to solve, compared to LP [105].

As mentioned earlier, by repeatedly adding columns, or variables, to the model, CG iteratively solves large-scale optimization problems. CG algorithm is computationally expensive, achieving near-optimal solutions, which can handle problems with large numbers of variables, like network flow or evacuation routing models. The primary goal is to identify the problem's optimal solution with the minimum possible variables [109].

There is presently a variety of diverse metaheuristic-based solution approaches available in the area of global optimization research on metaheuristics. Particle swarm optimization, ant colony optimization, genetic, simulated annealing, and tabu search are among the first groups of classical metaheuristic algorithms. Metaheuristics can perform exceptionally well in situations where classical or traditional exact optimization methods would fail to generate satisfactory results, in addition to scaling perfectly well in finding high-quality solutions to many complex and NP-hard problems (nondeterministic polynomial time-hard problems). The ability of metaheuristic problem-solving techniques to produce high-quality and robust solutions in comparatively shorter amounts of time than their exact optimization approach counterparts is certainly one of their main advantages. However, metaheuristic techniques do not produce optimal solutions such as IP and LP, instead, near-optimal solutions will be generated [107].

Generally speaking, the concept of simulation is to mathematically replicate a real-world scenario, analyze its operational features and attributes, and then establish assumptions and take suitable measures based on the simulated outcomes [105]. Large-scale, intricate real-world problems that are unsolvable with traditional decision models (e.g., IP) can be analyzed through simulation. Developing and solving a purely mathematical model of a municipal government system, for instance, might not be able to adequately account for significant political, social, economic, and environmental issues. However, simulation is capable of successfully modeling such systems. In a short amount of time, a computer simulation model may capture the results of scheduling, publicity, or other regulations throughout several months or years. On the contrary, developing an efficient simulation model can be highly costly, and the process is frequently long and complex. Compared to other decision modeling methods like IP or LP, simulation fail to deliver optimal solutions to problems. Also, each model of a simulation is distinct. Most of the time, its conclusions and solutions cannot be applied to different situations [105].

There are several uncertainties generators during evacuation, such as human behavior and time, place, and magnitude of the catastrophe, which are overlooked in the optimization literature, and the focus on deterministic solutions. Neglecting to consider these uncertainties may lead to catastrophic outcomes concerning people's safety. Hence, researchers should solve stochastic and robust evacuation planning models. Even though static evacuation problems can be applied to large-scale networks, they cannot represent the dynamics of evacuation statistically. The optimization problem of realistic size can be solved with dynamic models only with the employment of heuristic methodologies; therefore, developing further approaches that can handle these kinds of issues because they are far more difficult to resolve than deterministic evacuation planning problems is a significant matter that should be addressed [21]. Gai et al. [158] suggested solving optimization problems regarding the evacuation route selection by considering the existence of uncertainties.

A great portion of the reviewed studies assume the evacuation process with private vehicles, although this is not the case, as many people are taking mass-transportation vehicles due to their incapacities; thus, mass-transit-based evacuation and multi-model evacuation should be investigated more. According to Bayram [21], the available research on multi-modal and mass-transit evacuation planning often examines optimal strategies to organize transit schedules, identify the best routes for picking up travelers, and determine the most suitable locations for pick-up during an emergency evacuation. Compared to the model where evacuation via only private vehicles is presumed, this results in a more complex evacuation problem to solve because the first model introduces vehicle routing challenges into the second. In light of this, heuristic approaches are typically employed to solve these problems.

Modeling mass evacuation with consideration to the public transport (e.g., busses) enhances a fast, economical, and smooth evacuation process within the available timeframe [159]. Assessing the risk has a great influence on determining the zones that are exposed to severe hazards and require compulsory evacuation; nevertheless, the present literature lacks risk assessment applications in traffic assignment, which needs further examination to enhance evacuation planning in transportation simulation. A tendency towards using private vehicle services (such as Uber and taxis) is so common these days; yet multi-modal evacuation studies did not cover the possibility of utilizing these means to assist in evacuating low-income residents, although the conclusions may not give optimistic results. Taking different commuting patterns into account can be challenging due to the incorporated dynamics of such systems, however, there is a lack of consideration of multimodal systems in evacuation research, while it is a substantial subject that should be addressed. Alaeddine et al. [85] emphasized the importance of safety, minimizing delays, and reducing overall travel time in the optimization of evacuation plans.

The dynamic nature of the disaster environment necessitates the immediate installation of urgent rescue facilities, such as gas sensors and evacuation indicator signs, in certain critical parts of the evacuation path. The emergency evacuation indicator system's wireless data transmission mode is a fundamental technological advancement that enables the intelligent large-scale evacuation of people in emergencies. According to the current circumstances at the site, an intelligent emergency evacuation system based on communication technology and evacuation decision optimization can direct personnel to avoid hazardous areas and crowded routes and assist all evacuees in leaving the accident site quickly and safely [160].

In summary, this research suggests the employment of dynamic traffic assignment to model evacuation routing while capturing a number of dynamic parameters (e.g., human behavior, congestion rate, and demand uncertainty). Moreover, traffic simulation software is useful in modeling traffic networks while accounting for the individual characteristics of evacuees.

10.3. Evacuation planning and management strategies

Disaster management authorities are the decision makers during natural/human-made disasters, and thus, their perceptions are vital for modeling human behaviors. One of the most useful and well-known approaches in deciding between alternatives and making optimal decisions is the optimization technique. Thus, this research aimed at highlighting the available optimization methods, with a focus on evacuation routing, that can assist in realizing enhanced and more realistic evacuation strategies with consideration to human behaviors. Decision models involve optimizing an objective function while satisfying the existing constraints to find the feasible/optimal solution [105]. Optimization approaches assist in finding the optimal solution that satisfies the constraints and maximizes/minimizes the objective function. Thus, they are widely used in decision support and choosing among many alternatives.

In planning for an evacuation, there will always be distinct planning, response, and accountability issues because of the political structure in which disasters occur [161]. China, the United States of America, India, the Philippines, Indonesia, Japan, Vietnam, Mexico, Bangladesh, and Afghanistan are the top ten nations by sub-groups of catastrophe occurrence from 2000 to 2019 [162]. The majority of natural disaster fatalities occur in developing countries [163], where a large population lives in inadequate housing located in vulnerable urban areas [164]. Every year, individuals are killed by building collapses because public construction is usually of poor quality [162]. In other circumstances, corruption and lack of competence take precedence over building codes [165]. Low and lower-middle-income countries are more likely to have insufficient governance, which is linked to inadequate disaster management and low resilience to natural disasters [166]. On the other hand, according to Shapira et al. [167], high-income nations like Japan and New Zealand have reduced the number of fatalities because they have improved seismic building rules and strengthened community readiness. But their financial losses were worse, coming in at 67 % [2]. Thus, employing optimization approaches assists disaster managers in low-income countries in making the best course of action with their limited resources. For example, deciding on shelter locations that maximize accessibility, effective debris management to minimize the cost of removal, planning evacuation routing to ensure the best capacity utilization of the traffic network while minimizing total evacuation time, and efficient distribution of the available supplies to ensure that they are delivered with no delays and reached the most vulnerable people.

By way of illustration, Atando and Sugawara [162] compared the income-based management methods. They investigated how the financial resources of the government tend to affect the flexibility of coming up with solutions to prevent reoccurring natural calamities. Results showed that **the low-income country has been able to prepare for severe earthquakes by utilizing the collective power of the national and local governments, as well as partnerships with foreign organizations, the private sector, and civil society organizations.** To attain a significant and developed level of earthquake preparedness, **the high-income country undertakes investments in innovation, logistics, providing incentives to the governed, and adaptability.** Overall, the integration of optimization approaches in resource allocation will result in robust and efficient evacuation plans that make better use of the available resources.

A national evaluation of evacuation procedures and plans in hurricane-prone states was carried out by Urbina and Wolshon [168] is a good illustration of how legal systems work during emergency evacuation. Researchers observed that states' preparations are customized due to their distinct transportation systems and geographic locations. States typically use a two-stage approach to disaster preparedness and response, with state-level organizations coordinating efforts and local entities managing planning, response, and recovery procedures. The majority of states give their governors the legal power to issue instructions to evacuate, but others assign the role to local officials because of their experience with the area and the politically delicate aspect of evacuations. With respect to the legal framework, optimization approaches (i.e., job assignment and resource allocation) can assist in promoting the system's resilience and effectiveness in mitigating disasters.

Building on this, it is critical to establish evacuation plans with consideration of cultural variations in the different areas, to enhance public compliance. This is evident in the case of behaviors in socialist nations like Taiwan are motivated by fundamental cultural values that are expressed in shared purpose and activity [157]. On the contrary, in nations like New Zealand that have an individualist culture, people behave consistently in all circumstances to reflect a self-concept that is mostly independent of social context and places a high value on attaining one's goals. Similarly, within Western nations, the likelihood that social context will impact individuals' perceptions of risk and their decisions about risk management is evidence of fundamentally individualistic cultures. When faced with uncertainty, individuals seek guidance from like-minded individuals to minimize inconsistency and make risk management decisions [158]. Conversely, public awareness of seismic hazards may have a greater impact on preparedness intentions than cultural considerations [169].

Bringing all these aspects together, evacuation routing planning is influenced by many factors. For example, a country's economic level determines the quality of the existing traffic network infrastructure. If there is a poor infrastructure (e.g., poor drainage system and building structures), it may cause great human and property damage. Further, high-income level countries can invest a large amount of money in research and development to enhance their preparedness system. Technological development in the areas where the disasters occur may influence the disaster outcomes. For example, installing sensors/devices within the traffic network that assist in risk assessment may lead to better risk mitigation.

To conclude, disaster management authorities are faced with many managerial decisions. For example, whether to delegate issuing evacuation orders to lower levels, as they would be more knowledgeable of local barriers including cultural attributes and geographical structure. Accounting for cultural differences (e.g., compliance rate) could be a vital matter in establishing successful evacuation plans. Further, **each area has its limited resource availability including manpower, financial, equipment, research and development laboratories, and so on.** Thus, authorities need to manage the existing resources effectively, as well as, seek to expand their re-

sources and improve their current capacities (e.g., shelters and evacuation routes), especially for the low- and moderate-income levels countries. Effective strategies include maximizing capacity/resource utilization and/or minimizing the total cost (e.g., time, budget, and effort) by employing optimization methods. As well as account for the human factors/behavior variabilities to ensure effective implementation and successful outcomes of the established evacuation plans.

11. Research gaps and future directions

After reviewing various studies, this review paper finds some limitations in the area of study as follows.

- **Less emphasis is made on how individuals behave during a vehicle evacuation;** most research focused on how people respond during a building evacuation [33]. **Future studies should address drivers' behavior,** i.e., how drivers would behave differently with the various evacuation signage systems, in the context of large-scale emergencies, such as floods, tsunamis, and volcanic evacuations.
- As mentioned earlier in Section 7.1.2., **there is limited research in studying vulnerable individuals,** i.e., people with chronic diseases, within the traffic network. AbdelMagid et al. [155] developed a conceptual framework that provides a formal method of incorporating the needs of vulnerable populations into the planning process to enhance the transport system's outcomes for vulnerable communities, with a focus on resilience to disasters. Future studies should integrate the outputs of empirical research investigating vulnerable individuals' behavior, hence, input the results to enhance the transport system's resilience, employing the abovementioned framework.
- There is limited research quantifying human factors/behavior and consequently inputting this data into optimization models. **Future studies should investigate human behavior such as compliance with evacuation routing** with driving simulator technologies, and then extract the numerical findings as inputs to optimize the evacuation routing by solving a stochastic optimization problem, employing traffic assignment models.
- **Large-scale evacuation contexts such as avalanches and heatwaves are research topics that did not get much attention from the present studies and should be explored further.**
- **Future studies should investigate the relevance between income level and the use of public transportation means.** Considering that lower-income individuals may have limited access to private vehicles and are more vulnerable to disasters (i.e., limited resources and inadequate construction). Such a study will assist in securing equality between the community members, as well as enhancing resilience to disasters by identifying areas where further support is necessary. Also, what factors influence evacuating with public transportation, and how these results could benefit the evacuation planners in attaining more successful evacuation plans.
- Another significant issue that has not been addressed in the present literature is implementing a comprehensive study that integrates studying human factors/behavior in both buildings and the traffic network, given that large-scale evacuation (e.g., flood and hurricane) is concerned with both aspects. Future investigation could address how people would comply with a phased flood evacuation order, and then, study their driving behavior (e.g., with the influence of the number of household members evacuating) in the traffic network. The results may give a holistic overview of a phased flood evacuation order, which could give insights to disaster managers in optimizing shelter locations and other managerial decisions.
- The different contexts of evacuation are, to some extent, adequately investigated, yet it has been observed that some contexts have not been addressed. For instance, **university campus evacuation needs to be studied** since there are different facilities (e.g. nurseries, medical centers, accommodation, and labs with chemical/flammable materials) with different behavioral conducts and populations (e.g. vulnerable individuals in medical centers and nurseries). An optimization of the evacuation routing from the campus to the main roads could be implemented with consideration of several stochastic parameters (e.g., congestion rate and roads' capacities) to enhance the evacuation model.
- **Industrial cities evacuation context is absent in the literature.** Industrial cities incorporate large plants that may contain flammable and hazardous materials. Future research should study the decision to evacuate from a plant within an industrial city as a result of a fire in one of the plants while accounting for the fire spread. This research could be significant since if the fire spreads and reaches hazardous materials in other plants could result in escalating the problem at a rapid pace. Also, the routes in industrial cities could be of a low capacity to handle all the employees, as well as lead them to the main roads. Thus, a small-scale evacuation routing optimization model can be solved too.
- **Countryside also did not appear in the literature;** their education levels, road familiarity, and evacuation knowledge could be interesting factors to be explored and considered in planning for evacuation.
- It seems that there is a shortage of research that considers charting the behavioral patterns as constraints within the optimization problem to solve the routing problems. Overlooking the dynamics of evacuation aspects generates underestimation/overestimation in the associated parameters (e.g., clearance time).
- **Destination choice is among the least examined factors** in the literature, although it represents a valuable input into evacuation routing. This is due to the complexity inherent in the dynamics of human preferences regarding their destination since many factors can influence the targeted secured area. For example, the presence of friends, relatives, or even family members who need to be collected for evacuation.
- Investigating the compliance rate of shadow evacuees and how this can affect shelter and road capacities, then, the safety of individuals who are in the affected areas since delays in the traffic flow and/or overcapacity in shelters may occur.
- **A few studies have focused on developing, upgrading, and updating computer software and platform features so that they can be integrated into evacuation planning with the existence of an appropriate dataset.** There is also a need for built-in functions to

transmit behavioral investigations into traffic assignment problems, which calls for the software/computer engineers' attention to conduct such research.

12. Conclusion

This study set out to bridge the gap between the investigation of human behaviors during emergencies and the optimization of evacuation routing. Therefore, a comprehensive systematic review was conducted to address all, to the best of the author's knowledge, the available literature that aims to examine, or at least point to, an interdisciplinary approach in evacuation planning spanning from January 2012 to April 2024. The reviewed studies confirmed that the decision to evacuate and route choice are the most investigated factors that can be predicted to input into evacuation decision support. This study also identified that ABS is a common technique for conducting behavioral studies. The third major finding was that heuristics/metaheuristics and traffic assignment were frequently used approaches to design/assess evacuation routing. Logit/decision modeling, SFM, and CA emerged as reliable predictors of human behavior patterns during emergencies. It is worth mentioning that the CA model cannot capture movement speed and direction, which makes it useless in investigating an individual's characteristics and heterogeneity. By contrast, SFM models can address the entity's movement, which is influenced by destination choice, interactions (with other entities and surroundings), and other attractive forces. Further, ABS offers the advantage of adopting a heterogeneous interpretation of the agents.

Overall, this study strengthens the idea that a selected and manageable number of human factors should be involved in evacuation planning. Similarly, reflecting the interaction between emergency responders and evacuees can be advantageous to design practicality.

The major limitation of this study is the timeframe reviewed can be extended to include other past years to reflect a wider range of evacuation planning methodologies. It employs two databases, which can be widened to expand the search range. The scope of reviewed publications was limited in terms of scale, in which small-scale (i.e., building) evacuations were excluded. Being limited to the mathematical modeling of evacuation routing, this study lacks the presence of optimization and simulation models that address decision support strategies regarding shelter location assignment, vulnerability assessment, relief distribution, search and rescue, debris management, restoration, etc. An issue that was not addressed in this study was whether each method's outputs could be further input into different methodologies, which is a vital part of integrating evacuation planning.

CRedit authorship contribution statement

Razan Y. Aldahlawi: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Conceptualization. **Vahid Akbari:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Glyn Lawson:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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